158.337 Database Development
Week 3 - Entity-Relationship (ER) Modeling

Read Chapter 4 and Chapter 5 (section 5.3 & 5.4)

This week we will focus on Entity-Relationship Modeling to understand which aspects of ERD must be considered for the design and implementation of the database. We will use the Crow's Foot notation to present the data model components (entities, attributes, etc.) in the ERD.

We will discuss many ERD concepts such as

- Relationship strength – identifying/ non-identifying
- Entities - weak/strong and existence dependent/independent
- Implementation of multi-valued attributes
- Composite entities
- Keys – primary, foreign, surrogate
- Connectivities and cardinalities
- Recursive relationships
- Derived attributes
- Sequences
- Flexible design cases

You may use either of the CASE tools -Visio or JDeveloper for modeling diagrams. The textbook uses Microsoft Visio Professional to create the ERDs. Note that ERDs are done at the conceptual level. However, MS Visio has an implementation focus. Therefore, M:N relationships are not directly supported. Instead, the designer is limited to modeling 1:1 and/or 1:M relationships. Although M:N relationships may properly be viewed in a relational database model at the *conceptual* level, *such relationships should not be implemented*, because their existence creates undesirable redundancies. Therefore, M:N relationships must be decomposed into 1:M relationships to fit into the ER framework.

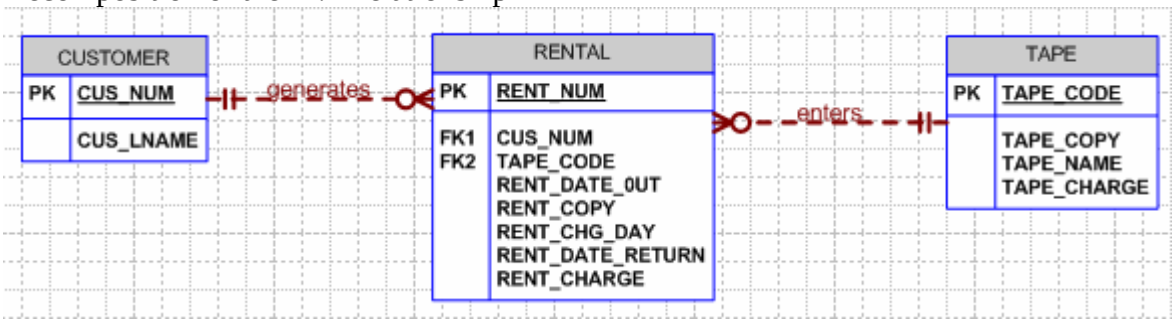
We also advise you to always decompose M:N relations into 1:M relationships. You will learn in this chapter that with decomposition, you can explicitly specify further design details (e.g., weak/strong entity, identifying/non-identifying relationships, keys, etc.)

For example, for a video rental store, you would note that tapes can be rented more than once and that customers can rent more than one tape.

1. A CUSTOMER may rent many TAPEs.
2. A TAPE can be rented by many CUSTOMERs.

Note the optional representation. Some customers do not rent tapes. Such customers might buy tapes or other items. Therefore TAPE is optional to CUSTOMER. Similarly, some TAPES are never rented. Therefore, CUSTOMER is optional to TAPE.

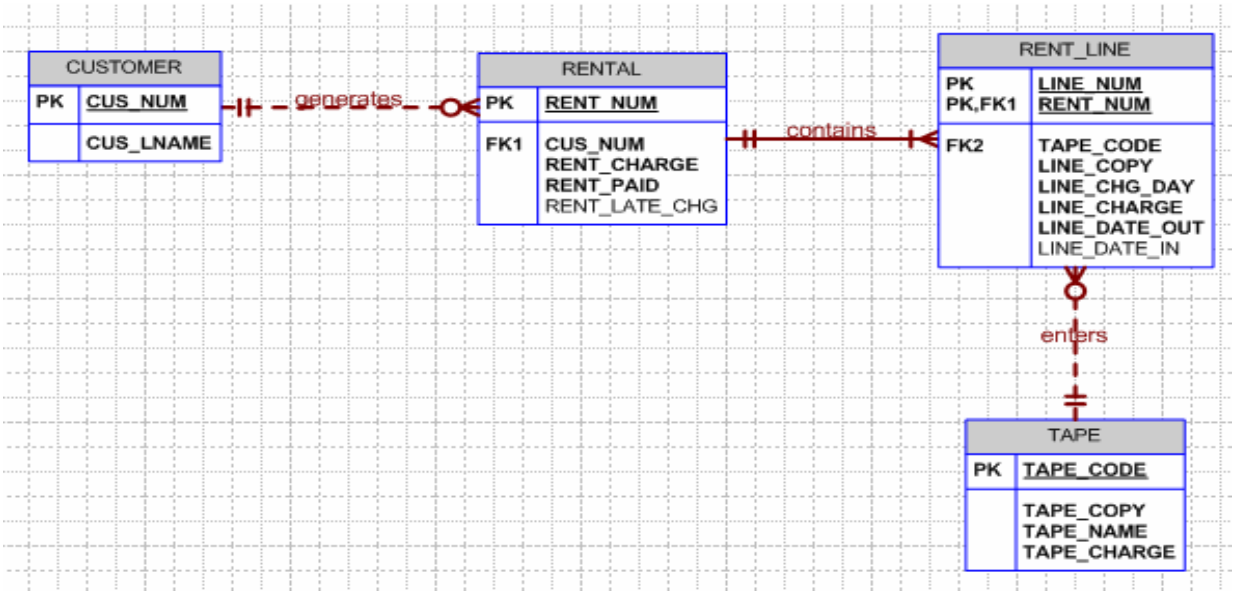
Decomposition of the M:N relationship



Remember that the relationships are read from the 1 side to the M side.
Therefore, the relationships between CUSTOMER and RENTAL and between TAPE and RENTAL are read as

1. CUSTOMER generates RENTAL
2. TAPE enters RENTAL

Further decomposition:



A completed ERD is the actual blueprint of the database. Its composition must reflect an organization's operations accurately if the database is to meet that organization's data requirements. ERDs form the basis for a final check on whether the included entities are appropriate and sufficient, the correctness of attributes, and the relationships between these entities. It is used as a final crosscheck against the proposed data dictionary entries. The ERD lets the designer communicate more precisely with those who commissioned the database design. Finally, the completed ER diagram serves as the implementation guide to those who create the actual database. In short, the ER diagram is as important to the database designer as a blueprint is to the architect and builder.

Chapter 5 contd….

Surrogate primary key - is an "artificial" PK that is used to uniquely identify each entity occurrence when there is no good natural key available or when the "natural" PK would include multiple attributes. A surrogate PK is also used if the natural PK would be a long text variable. The reason for using a surrogate PK is to ensure entity integrity, to simplify application development – by making queries simpler – to ensure query efficiency – for example, a query based on a simple numeric attribute is much faster than one based on a 200-bit character string -- and to ensure that relationships between entities can be created more easily than would be the case with a composite PK that may have to be used as a FK in a related entity.

Further, when implementing a 1:1 relationship

Case	ER Relationship Constraints	Action
I	One side is mandatory and the other side is optional.	Place the PK of the entity on the mandatory side in the entity on the optional side as a FK and make the FK mandatory.
II	Both sides are optional.	Select the FK that causes the fewest number of nulls or place the FK in the entity in which the (relationship) role is played.
III	Both sides are mandatory.	See Case II or consider revising your model to ensure that the two entities do not belong together in a single entity.

Time variant data - As the label implies, time variant data are time-sensitive. For example, if a university wants to keep track of the history of all administrative appointments by date of appointment and date of termination, you see time-variant data at work.

Design Trap - A design trap occurs when a relationship is improperly or incompletely identified and therefore, it is represented in a way that is not consistent with the real world. The most common design trap is known as a *fan trap*. A fan trap occurs when you have one entity in two 1:M relationships to other entities, thus producing an association among the other entities that is not expressed in the model.

NOTE: The tutorial exercise will be using many of these concepts.